

TrES-3b

R-1839

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_150425 · created 2026-08-03T06:46:09+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457137.89418 +/- 0.001 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.146 +/- 0.024
Transit depth (Rp/Rs)^2	2.12 +/- 0.7 [%]
Orbital Inclination (inc)	82.37 +/- 0.71
Airmass coefficient 1 (a1)	1.04 +/- 0.01
Airmass coefficient 2 (a2)	-0.034 +/- 0.014
Scatter in the residuals of the lightcurve fit is	0.97 %
Best Comparison Star	#1 - [559, 342]
Optimal Aperture	3.88
Optimal Annulus	8.04
Transit Duration (day)	0.0571 +/- 0.0068

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.146 ± 0.024 vs 0.16309 (deviation 0.71σ)
Duration fit vs published	0.0571 d vs 0.0567 d (deviation 0.06σ)
Mid-time O–C	2.2 min
Depth SNR	6.1
Residual scatter	0.97 %

Light curve

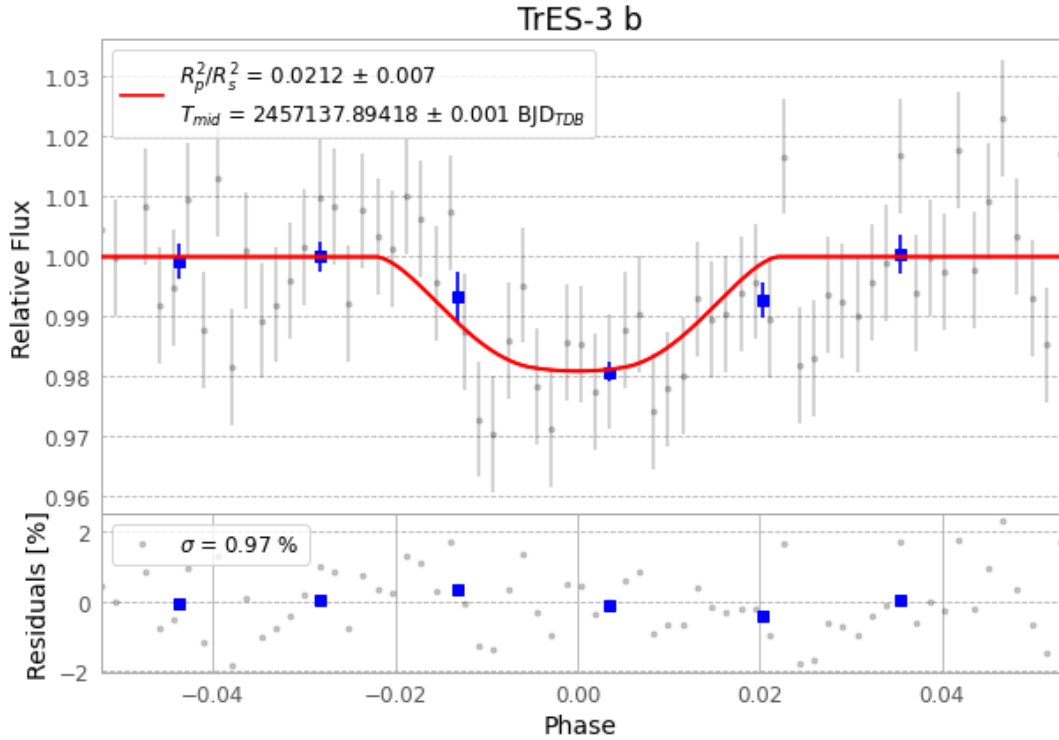


Figure 1 — FinalLightCurve_TrES-3 b_2015-04-25.png (SHA-256 35174f93922efbae...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [238, 258], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d7913c7519a84939...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

67 FITS frames (44 MB), TRES-3150425074512.FITS → TRES-3150425110312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-150425.log (6e708e259767...), FinalLightCurve_TrES-3 b_2015-04-25.png (35174f93922e...), FinalLightCurve_TrES-3 b_2015-04-25.csv (4f93f0adfc48...)

Compute resources

Wall time 170.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 06:46Z.